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Problem

- Strong reasoning models often depend on extensive supervised chain-of-thought demonstrations created by humans or stronger models.
- The central question is whether correct-answer rewards alone can make advanced reasoning patterns emerge.

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Distillation

- Outputs from DeepSeek-R1 fine-tune dense Qwen and Llama models from 1.5B to 70B parameters.
- Distillation works better than applying the same large-scale RL recipe directly to small base models.

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R1-Zero

PURE RL FROM A BASE MODEL

- Group Relative Policy Optimization compares sampled responses within a group, avoiding a separate value model.
- Accuracy and format rewards trigger longer reasoning, self-verification, reflection and an observed 'aha' transition.

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Evidence

MATH, CODE AND STEM

- DeepSeek-R1 reports 79.8% pass@1 on AIME 2024, 97.3% on MATH-500 and a 2029 Codeforces rating.
- The distilled 32B model surpasses OpenAI o1-mini on several reported reasoning benchmarks.

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R1 pipeline

COLD START + TWO RL STAGES

- A small curated long-CoT set improves readability before reasoning-focused RL on math, code and logic.
- Rejection-sampled reasoning plus general data support SFT, followed by RL for helpfulness, harmlessness and reasoning.

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Meaning & limits

- The work shows outcome-based RL can induce adaptive strategies without labeled reasoning paths and can teach smaller models.
- Long traces raise inference cost; reward hacking, language mixing, faithfulness, benchmark concentration and safety remain open.

DEEPSEEK-R1

Reasoning via Reinforcement Learning

DeepSeek-AI | arXiv:2501.12948v2

Use reinforcement learning on verifiable outcomes to elicit long-form reasoning, then combine cold-start data, broader alignment and distillation for practical models.

reasoning RLGRPOdistillation